Shinnosuke Yagi

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EDUCATION

Stanford University

M.S. in Statistics *GPA 3.86* — *Expected June 2026*

B.S. in Data Science *GPA 3.99* — *June 2025*

Relevant Coursework: CS 229 (Machine Learning), CS 246 (Mining Massive Datasets), STATS 100 (Mathematics of Sports), STATS 209 (Causal Inference), STATS 271 (Applied Bayesian Statistics), STATS 362 (Monte Carlo)

EXPERIENCE

Research Assistant – Biostatistics — **Stanford University**

Department of Statistics (Prof. Julia Palacios) — *September. 2024 - present*

- Applying a novel F-matrix-based distance metric for phylogenetic trees to improve detection of natural selection signals along chromosomes, leveraging tree topology and branch length information within a Hidden Markov Model framework.
- Optimized simulation processes, reducing runtime by 32%, and streamlined data conversion for statistical analysis in R.

Data Science Intern — **Austin, TX**

Zello — *June. 2025 – September. 2025*

- Developed an LLM-based pipeline to disentangle overlapping voice conversations from frontline worker communication, improving clustering accuracy (ARI from 0.29 to 0.91) and enabling the public release of Zello's AI Insights feature.
- Designed and maintained marketing and product analytics dashboards in Mode / Omni; built and tested dbt models with incremental strategies; enabled KPI tracking and A/B test analysis to measure feature impact on user engagement.

Data Science Intern — **Menlo Park, CA**

eneye — *August. 2024 – September. 2024*

- Architected AWS S3, RDS, and Lambda pipelines, optimizing 3D point cloud data processing by 57%.
- Developed and deployed a Dash-based web application on EC2 for monitoring of 3D space temperature trends.

Macro / Equity Research Intern — **Tokyo, Japan**

J.P. Morgan Chase & Co — *June. 2024 – August. 2024*

- Conducted quantitative research on U.S. / Japan macroeconomic policy and long-term interest rates using regression models on Fed Fund futures and Treasury spreads, delivering insights to senior economists and traders.

Research Assistant – Computational Social Science — **Stanford University**

Department of Sociology (Prof. Kiyoteru Tsutsui) — *March. 2022 – September. 2022, June. 2023 – December. 2023*

- Applied NLP libraries (Scikit-learn, Mecab) to analyze government publications, uncovering economic policy trends.
- Assembled and visualized the Varieties of Democracy (V-Dem) index for 200+ regions over 100 years using R, contributing to a published report in *Nikkei*, the world's largest financial paper.

PROJECTS

Machine Learning Optimization of Di-Higgs Signal-Background Separation in Collider Physics — *March. 2025*

- Increased di-Higgs signal significance by 20% (8.01→9.58) at XCC using neural network–BDT ensembles and genetic algorithms, enhancing Higgs self-coupling measurement precision in collider physics research.

Post-Wildfire Landslide Prediction *(Big Earth Hackathon Wildland Fire Challenge)* — *April. 2023 – June. 2023*

- Built a landslide prediction system using random forest model trained on rainfall forecasts and wildfire history, achieving 86% recall (64/74 successful predictions on actual 2023 post-wildfire landslides).

AWARDS **Toshin Scholar** (Full-ride $475k merit scholarship from Nagase Brothers Inc., Grand Prize among 40,000 students.)

SKILLS Python, R, SQL, C; NumPy, Pandas, scikit-learn, PyTorch, Tidyverse; AWS (EC2, S3, RDS, Lambda, IAM), Snowflake, dbt; Docker, Git; Tableau, Mode, Omni; ArcGIS Pro, QGIS; HubSpot, Jira, Confluence, Pipedream; Japanese (Native)